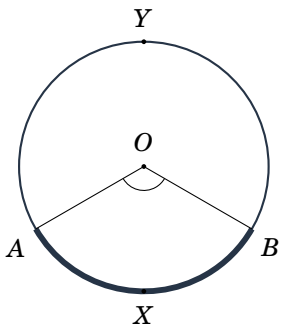
# Circles

Class 3

## 1 Arcs

Two points $A$ and $B$ on a circle cut it into two **arcs**. The smaller one is the **minor arc** and the larger one is the **major arc**. If $AB$ is a diameter, the two arcs are equal; each is a **semicircle**.




The angle that an arc **subtends at the centre** is the angle swept as the radius turns from $OA$ to $OB$ along that arc. The minor arc $AXB$ subtends $\angle AOB$. The major arc $AYB$ subtends the reflex angle $AOB$, which is $360° - \angle AOB$.

**Solve first.**

**P1.** In a circle with centre $O$, $\angle AOB = 100°$.

(a) Which arc subtends $100°$ at the centre: the minor arc or the major arc?

(b) What angle does the other arc subtend at the centre?

(c) What angle does a semicircle subtend at the centre?

## 2 The angle at the centre and the angle on the circle

The angle that arc $AB$ **subtends at a point** $C$ on the rest of the circle is $\angle ACB$. In P2 to P4, $O$ is the centre and $CD$ is a diameter. Draw a rough diagram for each and mark every angle you find.

**Solve first.**

**P2.** $A$ is a point on the circle, and $\angle OCA = 25°$.

(a) Why is $\angle OAC = 25°$?

(b) $\angle AOD$ is an exterior angle of $\triangle OAC$. Find it.

(c) Compare $\angle AOD$ with $\angle ACD$.

**P3.** $A$ and $B$ lie on opposite sides of the diameter $CD$, with $\angle OCA = 20°$ and $\angle OCB = 30°$.

(a) Use P2 to find $\angle AOD$ and $\angle BOD$.

(b) Find $\angle AOB$ and $\angle ACB$.

(c) Compare $\angle AOB$ with $\angle ACB$.

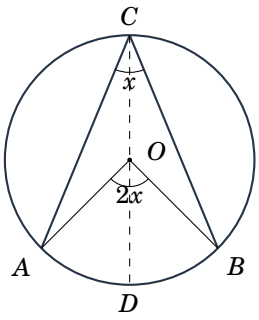
**P4.** $A$ and $B$ lie on the same side of the diameter $CD$, with $\angle OCA = 40°$ and $\angle OCB = 15°$.

(a) Find $\angle AOD$ and $\angle BOD$.

(b) Find $\angle AOB$ and $\angle ACB$.

(c) Compare $\angle AOB$ with $\angle ACB$. What pattern do P2, P3 and P4 show?

**Theorem 9** (NCERT Theorem 9). *The angle subtended by an arc at the centre of the circle is double the angle subtended by the arc at any point on the circle outside the arc.*



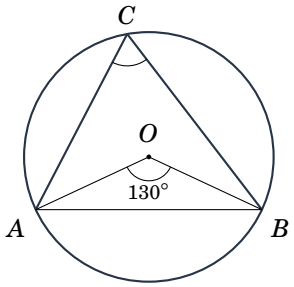

**Why it works.**

**Given:** arc $AB$ of a circle with centre $O$; $C$ is a point on the circle outside the arc.
**To show:** $\angle AOB = 2\angle ACB$.

1. Draw the diameter $CD$. $OA = OC$ (radii), so $\angle OAC = \angle OCA$. The exterior angle of $\triangle OAC$ is $\angle AOD = \angle OAC + \angle OCA = 2\angle OCA$.
2. In the same way, $\angle BOD = 2\angle OCB$.
3. If $O$ lies inside $\angle ACB$, add: $\angle AOB = \angle AOD + \angle BOD = 2(\angle OCA + \angle OCB) = 2\angle ACB$.
4. If $O$ lies outside $\angle ACB$, subtract: $\angle AOB = \angle AOD - \angle BOD = 2(\angle OCA - \angle OCB) = 2\angle ACB$.

The same steps work when the arc is a major arc; then $\angle AOB$ is a reflex angle.

**Example 1.** In a circle with centre $O$, $\angle AOB = 130°$, and $C$ is a point on the major arc $AB$. Find $\angle ACB$ and $\angle OAB$.




**Given:** $\angle AOB = 130°$; $C$ is on the major arc.
**To find:** $\angle ACB$ and $\angle OAB$.

1. $C$ is outside the minor arc $AB$, which subtends $130°$ at $O$. So $\angle ACB = \dfrac{130°}{2} = 65°$ (Theorem 9).
2. $OA = OB$ (radii), so $\angle OAB = \angle OBA = \dfrac{180° - 130°}{2} = 25°$.

**Answer:** $\angle ACB = 65°$ and $\angle OAB = 25°$.

## 3 Two consequences

**Solve first.**

**P5.** In a circle with centre $O$, $\angle AOB = 80°$. $C$, $D$ and $E$ are three points on the major arc $AB$.

(a) Find $\angle ACB$, $\angle ADB$ and $\angle AEB$.

(b) Does the answer depend on where $C$ is on the major arc?

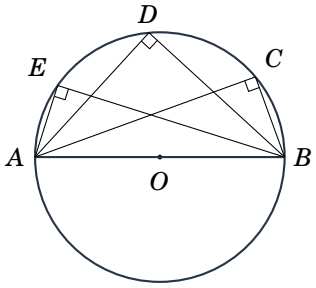
**P6.** $AB$ is a diameter of a circle with centre $O$, and $C$ is any other point on the circle.

(a) What angle does the semicircle not containing $C$ subtend at $O$?

(b) Find $\angle ACB$.

The part of the circle on one side of a chord, together with the chord, is a **segment**. P5 shows that *angles in the same segment are equal*: each is half the same angle at the centre.

**Corollary of Theorem 9** (NCERT Corollary). *The angle subtended by a diameter at any point on the circle is* 90°.




**Why it works.**

**Given:** $AB$ is a diameter of a circle with centre $O$; $C$ is any other point on the circle.
**To show:** $\angle ACB = 90°$.

1. The semicircle not containing $C$ subtends a straight angle, 180°, at $O$.
2. By Theorem 9, $\angle ACB = \dfrac{180°}{2} = 90°$.

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**Things to Remember**

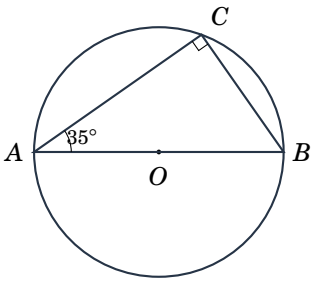
The angle at the centre is double the angle on the circle, for the same arc.
Angles in the same segment are equal.
The angle in a semicircle is a right angle.

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**Solve first.**

**P7.** Draw a circle and a chord $AB$. Mark a point $C$ on the major arc, a point $P$ inside the circle and a point $Q$ outside it, all on the same side of $AB$. Measure $\angle ACB$, $\angle APB$ and $\angle AQB$. Which is the largest? Which is the smallest?

**Example 2.** $AB$ is a diameter of a circle with centre $O$, and $C$ is a point on the circle with $\angle CAB = 35°$. Find $\angle ABC$.




**Given:** $AB$ is a diameter; $\angle CAB = 35°$.
**To find:** $\angle ABC$.

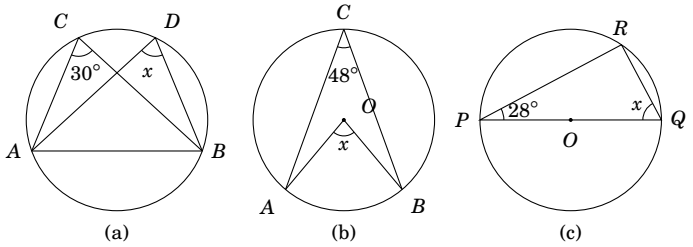
1. $\angle ACB = 90°$ (angle in a semicircle).
2. $\angle ABC = 180° - 90° - 35° = 55°$ (angle sum of a triangle).

**Answer:** $\angle ABC = 55°$.

Draw a rough figure for every question in Exercise 3A that has no figure.

## Exercise 3A

1. An arc of a circle subtends an angle of $70°$ at the centre. Find the angle it subtends at a point on the rest of the circle.
2. In a circle with centre $O$, $C$ is a point on the major arc $AB$ and $\angle ACB = 40°$. Find $\angle AOB$ and $\angle OAB$.
3. In a circle with centre $O$ and radius 12 cm, $\angle AOB = 60°$. Find the length of the chord $AB$, and the angle that $AB$ subtends at a point on the major arc.
4. In a circle with centre $O$, $\angle AOB = 100°$, and $D$ is a point on the *minor* arc $AB$. Ravi writes $\angle ADB = 50°$. Find his mistake, and find $\angle ADB$.
5. $AB$ is a diameter of a circle and $C$ is any other point on the circle. Find $\angle ACB$ and explain your reasoning without using Theorem 9: join $OC$ and use two isosceles triangles.
6. Find $x$ in each figure. $O$ is the centre.




7. A carpenter wants to check that a wooden piece has a true semicircular edge. She places the corner of a set square on the curved edge so that its two edges pass through the ends of the straight edge. She repeats this at several places. Explain the test.

8. $MNOP$ is a quadrilateral inscribed in a circle, and $MN$ is a diameter. What can you say about $\angle MOP$ and $\angle MNP$? Explain.

9. $A$ and $B$ are points on a circle.

   (a) Can there be points $X$ and $Y$ on the circle, on the same side of $AB$, with $\angle AXB \neq \angle AYB$?

   (b) If $X$ and $Y$ are on the circle and $\angle AXB = \angle AYB$, must $X$ and $Y$ be on the same side of $AB$?

   (c) If $\angle AXB = \angle AYB$, and $X$ and $Y$ are not on the circle, does the circle through $A$, $B$ and $X$ also pass through $Y$?

10. In Class 2 we saw by construction that the circumcentre of a right-angled triangle is the midpoint of its hypotenuse. Prove it. Let $\triangle PQR$ have $\angle P = 90^\circ$ and circumcentre $O$, and use Theorem 9 for the arc $QR$ that does not contain $P$.

11. A chord of a circle is equal to the radius. Find the angle it subtends at a point on the major arc.